Integrated Life Cycle Sustainability Assessment of a Hybrid Biochemical – Thermochemical Pathway for Slaughterhouse Waste Valorisation

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Abstract: Slaughterhouse waste (SHW) and animal by-products (ABPs) possess considerable potential for renewable energy generation due to their high organic matter content. Within the framework of the international research project BIOTHEREP ("Hybrid Biochemical and Thermochemical Conversion of Slaughterhouse Biowaste for Renewable Energy Production"), an integrated biochemical–thermochemical valorisation system was developed and evaluated regarding its environmental and economic performance. The assessment was conducted using Life Cycle Assessment (LCA), Material Flow Cost Accounting (MFCA), as well as sensitivity and uncertainty analyses. The results demonstrate significant environmental advantages compared to conventional disposal pathways. On average, greenhouse gas emissions were reduced by 188.2 kg CO_2-eq. per ton of slaughterhouse waste. The highest environmental benefits were observed in South Africa (−460.6 kg CO_2-eq./t) and Morocco (−296.5 kg CO_2-eq./t). Positive effects were also identified with respect to fossil resource depletion, agricultural land occupation, and eutrophication-related impact categories. The sensitivity analysis confirmed the robustness of the results across different weighting approaches. In contrast, the economic assessment revealed that profitability strongly depends on site-specific energy prices and regulatory conditions. Although the overall cost structures were comparable across countries, considerable differences were observed in revenue generation. Overall, the findings highlight the potential of integrated bio- and thermochemical conversion systems to support the sustainable utilisation of animal by-products. At the same time, they emphasize the importance of appropriate market incentives and policy frameworks to enable the large-scale implementation of such technologies.

Keywords: Slaughterhouse waste, Circular economy, Life Cycle Assessment, Material Flow Cost Accounting, Biogas, Biochar, Anaerobic digestion, Sustainable Analysis

1 Introduction

Against the backdrop of increasing resource scarcity, rising energy demand, and ambitious climate protection targets, the development of circular value creation systems is gaining growing importance. Organic waste streams are increasingly being regarded as valuable resources for renewable energy production and the recovery of secondary raw materials. Animal by-products and slaughterhouse waste occupy a special position in this context. They are generated in large quantities worldwide along the meat production chain and possess considerable energy potential due to their high content of proteins, fats, and organic matter. At the same time, their safe storage, transportation, and treatment pose significant challenges because of strict hygienic requirements. Conventional disposal pathways, such as incineration or rendering, often result in the loss of valuable carbon and nutrient resources and are associated with additional environmental burdens. To improve resource efficiency and promote the closure of material cycles, an integrated valorisation system combining biological and thermochemical conversion processes was developed within the framework of the international research project BIOTHEREP. To evaluate the environmental and economic performance of this system, an integrated assessment approach was applied, combining Life Cycle Assessment (LCA) and Material Flow Cost Accounting (MFCA). In addition, sensitivity and uncertainty analyses were conducted to assess the robustness of the results under different modeling assumptions.

2 Related Work

The energetic use of animal by-products and slaughterhouse waste has been investigated for several years as a promising approach to increasing resource efficiency and providing renewable energies. The focus of research to date has been on the anaerobic digestion of organic residues to produce biogas. Numerous studies have shown that animal by-products enable high methane yields due to their high content of proteins and lipids and thus have a relevant potential for substituting fossil fuels [1–3]. Several life cycle assessment studies demonstrate the ecological advantages of biogas production from animal residues. Among other things, significant reductions in greenhouse gas emissions and fossil resource consumption compared to conventional disposal methods have been demonstrated [4, 5]. At the same time, these studies identify methane emissions, the treatment of digestate and the provision of process energy as key environmental hotspots. In addition, techno-economic analyses show that the profitability of such plants depends heavily on energy prices, subsidy mechanisms and the availability of suitable substrates [2, 6]. Despite extensive research, the majority of studies to date have focused on individual process steps in the recycling chain, in particular anaerobic digestion. Thermochemical post-treatment processes such as pyrolysis or gasification have so far usually been considered separately. Likewise, ecological and economic assessments are often carried out independently of each other. Holistic analyses of integrated systems that combine biological and thermochemical conversion processes and at the same time take ecological and economic aspects into account are only available to a limited extent so far. [2, 7] In addition, there is a lack of comparative studies that analyze the performance of such systems under different regional conditions. In particular, the influence of national energy systems, market structures and recycling options on ecological and economic sustainability has not been sufficiently investigated so far. Existing studies focus predominantly on individual countries or regions and therefore allow only limited statements on the transferability of the results to different socio-economic and energy contexts [4, 5]. The identified research gaps relate in particular to the lack of linking biological and thermochemical recycling processes as well as the limited integration of ecological and economic assessment approaches. The present study as part of the BIOTHEREP research project therefore examines the sustainability performance of an integrated valorisation system for SHW under different regional conditions. Through the combined application of LCA and MFCA, both environmental impacts and cost structures along the entire process chain are analyzed. The cross-country comparison also enables an assessment of the influence of national energy and market conditions on the performance of the system.

3 Material and Methods

As As part of the BIOTHEREP research project, an integrated valorisation system for slaughterhouse waste and animal by-products was developed that combines biological and thermochemical conversion processes. After pre-treatment and anaerobic digestion of the substrates, the biogas produced is used to provide electricity, heat or biomethane. In addition, the carbon dioxide captured during gas upgrading is methanized together with hydrogen to increase the carbon efficiency of the system. The remaining digestate is then thermochemically treated by pyrolysis and gasification, generating additional energy and material flows as well as biochar.The ecological assessment was carried out using LCA in accordance with DIN EN ISO 14040 and DIN EN ISO 14044. The aim was to quantify the environmental impacts of the developed valorisation system in the participating partner countries. The functional unit was defined as one ton of slaughterhouse waste. The system boundaries include the provision and pre-treatment of slaughterhouse waste, anaerobic digestion, gas treatment, methanation and thermochemical treatment of digestate. Slaughterhouse waste was modelled as a waste stream with no upstream environmental impacts. The integrated valorisation system was modelled with the help of the life cycle assessment software

Umberto, which enables the integrated mapping of complex material and energy flows. All process steps were modelled and mathematically linked with each other. The foreground inventory data build on an initial comparative LCA of biogas-based slaughterhouse waste valorisation versus conventional thermal disposal, which was subsequently revised and extended to enable a systematic comparison across the seven BIOTHEREP partner countries. Background processes were consistently modelled using ecoinvent 3.6 as the standard background database, including country-specific electricity grid mixes and energy carrier data. LCA data availability shows comparatively limited heterogeneity between countries, as the ecoinvent database provides a unified modelling framework for background processes. The impact assessment covers the categories Global Warming Potential (GWP100), Fossil Resource Depletion (FRD), Water Depletion Potential (WDP), Freshwater Eutrophication Potential (FEP), Marine Eutrophication Potential (MEP) and Agricultural Land Occupation Potential (ALOP). In addition to the comparative overall assessment, a normalized and weighted single-score indicator was calculated. The data-based evaluation of the modelling results as well as the statistical analyses were carried out using the open-source programming languages R and Python (via Jupyter Notebooks). This included the calculation of the single-score indicators, the implementation of the sensitivity and Monte Carlo analyses, and the creation of the graphical evaluations and visualizations. By combining material flow modelling, life cycle databases and data analysis methods, an integrated environmental informatics assessment approach was implemented that considers both ecological and economic sustainability aspects. The economic evaluation was carried out using MFCA in accordance with DIN EN ISO 14051, also using the Umberto software. The analysis is based on a reference plant with an electrical output of 500 kW and an annual processing capacity of 10,000 t SHW. The economic input data were gathered in 2024 and 2025 through systematic desk research on country-specific energy prices, feed-in tariffs, and prevailing market conditions. These data exhibit considerably higher uncertainty and cross-country variability than the LCA inventory data, reflecting differences in market maturity and data availability between European and African partner countries. This uncertainty is methodologically accounted for through a two-scenario approach (with and without SHW input costs). In addition to investment and operating costs, material, energy and loss costs along the process chain were recorded. Slaughterhouse waste was valued at 300 €/t based on its opportunity cost. In addition, methane and hydrogen losses were considered in monetary terms. The revenue side includes revenues from electricity and heat generation, biomethane supply and potential by-products.

4 Results

The technical evaluation of the BIOTHEREP system confirms the fundamental suitability of slaughterhouse waste for integrated energy and material recycling.

Process modelling showed that approximately 210 m³ of biogas can be produced from one ton of slaughterhouse waste. This corresponds to an energy potential of around 1.8 MWh per ton of input material. The biogas yields achieved are in the upper range of comparable organic residue flows and illustrate the high energy potential of animal by-products. Slaughterhouse waste has high levels of proteins and lipids, which enable intensive microbial conversion during anaerobic digestion and thus promote above-average energy yields. The results thus confirm earlier studies that characterize animal by-products as a particularly energy-rich biomass fraction. At the same time, it was shown that the composition of the substrates has a significant influence on process stability. During fermentation, increased concentrations of nitrogen-containing compounds are produced due to the degradation of protein-rich components. The resulting ammonia formation is a central factor influencing the activity of methanogenic microorganisms. In addition, the high sulphur content of individual fractions led to the formation of hydrogen sulphide in the raw biogas, which creates requirements for downstream gas upgrading. Despite these challenges, stable plant operation could be ensured through suitable operating strategies and treatment steps. The thermochemical post-treatment of the solid digestate opens up additional material and energy potential. Even after biological conversion, the digestate continues to contain relevant amounts of organic matter that can be used by pyrolysis and gasification. This process generates biochar with a pronounced pore structure and a high specific surface area. Tests on biogas purification indicate that the biochar produced can be used for the adsorption of hydrogen sulphide and thus makes an additional contribution to improving gas quality. In addition, the energy-rich gas fractions produced during thermochemical treatment allow process heat to be provided within the system. As a result, energy-intensive process steps can be partially supplied by internally generated energy.

4.1 Assessment of environmental impacts

The clearest advantages of the examined system can be seen in the GWP 100 impact category. While the thermal treatment of animal by-products causes positive emissions, the use in the BIOTHEREP system leads to a negative net climate balance due to the substitution of fossil fuels. The analysis of the material flows showed that the thermal disposal of slaughterhouse waste causes emissions of about 54.68 kg CO_2-eq./t material. In contrast, the developed valorisation system enables a net saving of about 200.93 kg CO_2-eq./t input material. This results in an overall climate advantage of around 146 kg CO_2-eq./t of treated slaughterhouse waste. The main reasons for this are the displacement of fossil electricity and heat generation using the biogas produced and the improved carbon utilisation as a result of integrated methanation. There are differences between the countries examined in terms of the level of greenhouse gas reductions. In South Africa and Morocco, whose energy systems continue to have a relevant share of fossil fuels, there is particularly high potential for savings.

The developed system also has clear advantages in the Fossil Resource Depletion (FRD) category. The results show that electricity production makes the largest contribution to reducing fossil resource consumption. The country-specific analysis makes it clear that the savings in fossil resources are particularly pronounced in regions with a high dependence on coal, oil or natural gas. In South Africa, for example, the use of biogas can make an important contribution to reducing dependence on coal-based energy systems. At the same time, it becomes clear that individual process steps continue to depend on external energy and auxiliary material inputs. These include the pre-treatment of substrates, gas treatment and various auxiliary materials for membrane and cleaning systems. These processes cause part of the remaining resource consumption and represent potential areas for optimization for future system developments.

The evaluation of the impact categories Water Depletion Potential (WDP), Freshwater Eutrophication Potential (FEP) and Marine Eutrophication Potential (MEP) illustrates the complexity of bio-based valorisation systems. While clear environmental benefits are achieved in the climate-related categories, more differentiated results can be seen in water- and nutrient-related categories. Part of the environmental impact is due to the use of water during the pre-treatment and hygienization of slaughterhouse waste. In addition, nitrogen and phosphorus flows within biological processes can potentially contribute to eutrophication effects. However, the results show that these burdens remain of minor importance compared to the environmental reductions achieved. Particularly relevant in this context is the use of liquid fermentation residues. Through their controlled treatment and subsequent use, nutrients can be circulated and potential environmental pollution reduced. Regional differences arise regarding water availability. In water-scarce regions of Africa, the efficient use of process water is more important than in many European locations.

The analysis of agricultural land use (ALOP) also shows positive results. Since slaughterhouse waste is a residue from meat production and no additional biomass production is required, there is no additional competition for land, in contrast to energy crop systems. While in Europe the competition between energy, food and feed production is increasingly being discussed, in many African regions questions of food security and sustainable land use are in the foreground.

Table 1 Total environmental impact per country when using the biogas plant

	GWP 100	ALOP	FRD	WDP	FEP	MEP
	kg CO_2-eq.	$m^2 \cdot a$	MJ_2-eq.	m^3 H_2O-eq.	kg P-eq.	kg N-eq.
Algeria	-201,3	-14,5	-192,8	-1,3	-0,002	-0,04
Egypt	-193,9	-14,7	-160,0	-1,2	-0,002	-0,06

France	124,8	-21,2	-1,3	-2,5	-0,003	-0,04
Germany	-200,9	-39,4	-83,6	-2,3	-0,059	-0,06
Italy	-88,7	-42,3	-82,1	-3,6	-0,010	-0,06
Morocco	-296,5	-19,9	-135,3	-1,5	-0,021	-0,08
South Africa	-460,6	-21,5	-179,0	-2,6	-0,057	-0,14
Mean	**-188,2**	**-24,8**	**-119,2**	**-2,1**	**-0,022**	**-0,07**

In addition to the comparative overall assessment of the environmental impacts, a normalized and weighted single-score indicator was calculated. The results serve exclusively for comparative classification and do not replace the impact category-specific interpretation. For each impact category, a country normalization was used, based on the magnitude of the arithmetic mean of the countries considered, to obtain dimensionless, comparable values. The weighting of the impact categories was based on the Environmental Footprint methodology of the European Commission [8]. The EF method explicitly provides that impact categories may be merged into an aggregated single score after normalization, but at the same time emphasizes the normative character of this step. Weightings therefore do not reflect scientific objectivity but are based on social and political priorities. Central references for this are the ILCD Handbook [9, 10] and the Environmental Footprint Category Rules (EFCR), in which the European Commission clarifies that there is no binding, universally applicable weighting rate. Instead, methodological guidelines are provided within which weightings are to be justified in a transparent, consistent and comprehensible manner. In particular, the global warming potential is highlighted as a central impact category, while resource scarcity, land use and eutrophication-related effects are also classified as highly relevant for a comprehensive environmental assessment. The results show clear differences in the overall environmental performance of the systems studied. The highest environmental benefits were observed for South Africa, followed by Germany and Morocco. These countries are characterized by consistently high environmental credits across several impact categories and achieve above-average results, especially in the categories of global warming potential, fossil resource consumption and eutrophication potential. South Africa's high environmental performance is due to the pronounced greenhouse gas reductions and the significant advantages in marine eutrophication. In contrast, Algeria, Egypt and Italy have a medium overall rating. Although these countries also achieve significant environmental benefits in individual impact categories, the aggregate results are more moderate overall due to lower reductions in other categories. France is the only country surveyed with a positive net greenhouse gas balance. This result is not a methodological outlier, but a systemic effect of an already largely decarbonized electricity mix, in which additional electricity demand does not generate any further climate benefit. This result highlights the importance of a multidimensional environmental assessment, as positive results in individual impact categories do not necessarily lead to an overall high environmental performance.

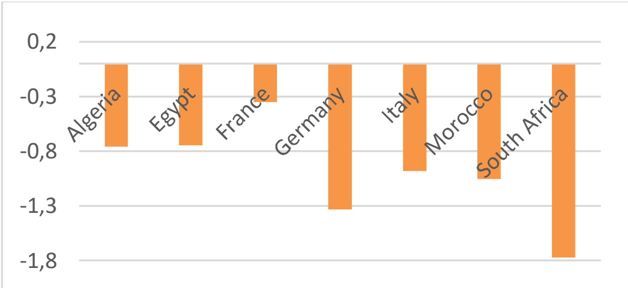

Fig. 1 Normalised environmental impacts by country

Sensitivity and uncertainty analyses

To assess the robustness of the results, deterministic sensitivity analyses and probabilistic uncertainty analyses based on Dirichlet-distributed weightings were performed. Different weighting schemes for the environmental indicators under consideration were examined and their influence on the aggregated environmental assessment was analyzed. First, the sensitivity to a changed weighting of the global warming potential was investigated. If the GWP weight is increased from 0.30 to 0.40 and the other categories are reduced proportionally, the benefits of countries with high climate-related savings are significantly increased. In this scenario, South Africa continues to improve to −1.8733 and remains in first place by a wide margin. Germany deteriorates slightly to −1.2337 but retains second place. Morocco stabilizes in third place with −1.1853. France reacts particularly clearly, whose single score deteriorates to −0.0964 and thus moves even closer to zero. This development confirms that France's additional climatic burden dominates the aggregate index under a more climate-political weighting. Despite these changes, the overall ranking remains unchanged, indicating a high structural robustness of the extreme positions. In the opposite scenario, in which the weight of the GWP is reduced to 0.20, a mirror-image effect is observed. France improves significantly to −0.3496, as the positive climate impact is less significant. Italy benefits more than average in this scenario and achieves a single score of −1.0460, which brings it closer to Morocco. South Africa remains in first place with −1.6820, while Germany remains stable in second place with −1.2891. Here, too, it can be seen that the reduction in climate priority changes the absolute distances, but does not cause a fundamental rearrangement of the ranking. In a further step, the sensitivity to the selection of individual impact categories was analyzed. If the WDP is completely excluded from aggregation and the remaining weights are renormalized, Italy's position in particular will change. Italy's single score deteriorates significantly to −0.8435, which means that Italy falls behind Algeria and Egypt in the ranking. This result illustrates that Italy's comparatively good position in the baseline scenario is largely based on high savings in water consumption. Countries such as South Africa (−1.8729), Germany (−1.2907) and Morocco (−1.1919), on the other hand, remain almost unchanged in their rankings. The exclusion of WDP thus shows that water-related effects are of great importance, especially for countries with a strong agricultural impact profile. A supplementary scenario considered the exclusion of the eutrophication-related categories FEP and MEP. In this case, Morocco improves slightly compared to Germany and achieves a slightly better value of −1.1382 than Germany (−1.1343). This reversal of ranking is due to the fact that Germany benefits disproportionately from high savings in the eutrophication-related categories in the baseline scenario. South Africa remains the clear leader in this scenario (−1.6482), while France continues to occupy the weakest position (−0.1903). The assessment in the upper midfield reacts sensitively to the consideration of individual impact categories, while the extreme positions remain stable.

Fig. 2 *Sensitivity analyses of normalised environmental impacts by country*

To further confirm the results, a probabilistic sensitivity test was carried out, in which the weights were randomly varied by the basic theorem, whereby all weights remained positive and always added up to one (cf. **Fig.** *2*). To meet these conditions, the Dirichlet distribution was used. This analysis showed that South Africa ranked first in all simulations and France remained in last place without exception. These results underline the high robustness of the central ranking relations against plausible weighting variations. When interpreting the score intervals, South Africa has the lowest (best) overall result with a median of −1.778. The interval [−1.947; −1.608] is completely below all other country intervals. There is no overlap with Germany or Morocco. This shows that South Africa has the highest aggregate savings potential among almost all plausible weighting combinations. The ranking position is therefore highly robust.

Germany follows with a median of −1.255 and an interval of [−1.413; −1.134]. This interval does not overlap significantly with either South Africa or Morocco. Germany thus remains stable in second place. The moderate interval range also shows that Germany's assessment is less sensitive to weighting changes than that of the countries in the midfield. Morocco has a median of −1.119 with an interval of

[−1.207; −1.035]. The interval boundaries are clearly above Germany and below Italy. Here, too, there is no relevant overlap with neighbouring countries. Morocco is thus robustly positioned in third place and consistently benefits from the greater weighting of climate- and resource-related categories. Italy has a median of −0.972, but the interval [−1.119; −0.835] is much wider. The partial overlap with both Morocco (at the lower end) and Algeria (at the upper end) is striking. This means that Italy's ranking position is not clear in all weighting scenarios. Depending on the weighting, Italy can move closer to Morocco or fall behind Algeria in certain areas. The country is thus a classic example of normative sensitivity. Algeria (median −0.838, interval [−0.952; −0.729]) and Egypt (median −0.810, interval [−0.898; −0.716]) show strongly overlapping intervals. This indicates that their relative ranking is not robust and depends largely on the concrete weighting assumptions. In terms of content, this means that both countries have very similar, moderately pronounced impact profiles, without clear dominance in a highly weighted category. France is a special case. With a median of −0.224 and an interval of [−0.401; −0.045], France is clearly separated from all other countries. Even the 5% quantile does not overlap with the 95% quantile of Egypt. This means that France is consistently the country with the lowest aggregate benefit among all plausible weightings. The proximity of the 95% quantile to zero also shows that France achieves almost no overall benefit in individual weighting combinations.

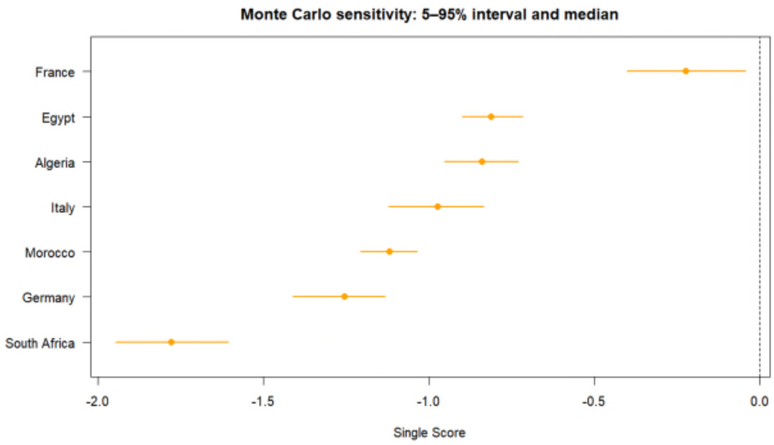

Fig. 3 Boxplot of Monte-Carlo-Simulation

4.2 Hotspot analysis of environmentally intensive and environmentally beneficial processes

Environmentally intensive processes

The analysis of the environmentally intensive processes showed a consistent pattern across all countries studied. The fermentation process had the highest aggregate contribution to the global warming potential, regardless of national differences in energy system, transport infrastructure or process design. The dominance of fermentation is mainly due to biogenic methane emissions from animal husbandry and confirms its role as a central environmental hotspot within the system under consideration. The assumed methane emissions of 1.75% are within a normal range for production. The nevertheless high environmental impact illustrates the importance of stable operation without leaks. Other stressful processes were identified, especially in the area of transport and the provision of electricity and heat. However, their contribution remained significantly lower compared to fermentation. In the North African countries of Algeria, Egypt and Morocco, the relative share of energy-related processes was slightly higher than in the European case studies. The reasons for this are the more fossil-fueled energy systems and lower efficiencies of upstream process chains. Nevertheless, fermentation remained the dominant burden driver in these countries as well. A comparable picture emerged for South Africa. The results make it clear that although technical optimizations in areas such as transport, energy efficiency or process integration can contribute to improvements in environmental performance, the fundamental environmental impacts are still largely determined by animal production itself.

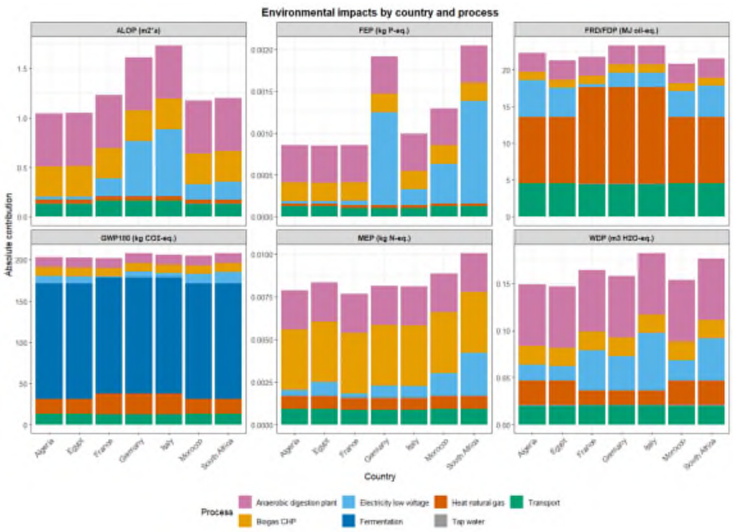

Fig. 4 Particularly environmentally intensive processes by country

Environmentally beneficial processes

In almost all countries studied, the greatest contribution to environmental relief was achieved by feeding electricity from biogas into the public power grid. In addition, credits from the substitution of mineral fertilizers contributed to improving the environmental balance. In Germany and Italy, electricity substitution was the dominant relief mechanism. The additional feed-in of renewable electricity displaces conventional power generation and thus leads to environmental credits in several impact categories. Fertilizer substitution also made a positive contribution, but remained significantly lower than the effects on the energy industry. France occupies a special position within the analysis. Although electricity substitution is also the most important relief process here, its contribution is significantly smaller than in the other countries. The reason for this is the already largely decarbonized French electricity mix, the displacement of which generates few additional environmental benefits. For Algeria, Egypt and Morocco, on the other hand, significantly higher relief effects were determined. The electricity systems there, which continue to be heavily influenced by fossil fuels, mean that even moderate amounts of substituted electricity result in considerable environmental benefits. These effects were particularly pronounced in South Africa. Due to the coal-based energy system, it achieved the highest environmental relief of all the countries studied.

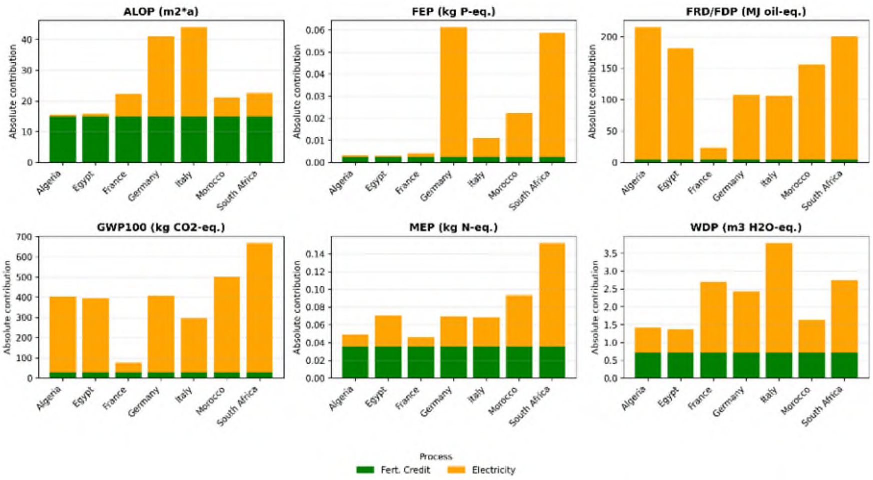

Fig. 5 Environmentally beneficial processes by country

4.3 Material flow cost accounting and profitability analysis

By integrating the MFCA into the existing life cycle model, physical material and energy flows could be directly linked to monetary variables. In addition to product flows, loss streams were valued in monetary terms in order to systematically make economic inefficiencies visible. The economic analysis first shows a remarkable stability of the cost structure between the countries under consideration. The cost structure is largely determined by the annuitized investment costs, fixed operating costs, and particularly the cost of the input material. Regional differences in energy and service costs, on the other hand, have only a limited impact on total costs. The technical cost structure can therefore be regarded as largely location-independent. In contrast, the revenue side shows considerable differences between countries. Product revenues vary from around 32 €/t input material in South Africa and France to over 120 €/t in Germany and Italy. These differences are mainly due to the respective electricity price levels and national support mechanisms. While the technical processes deliver identical product yields, the market value of the energy generated largely determines the economic performance of the system. A key finding of the MFCA is the importance of loss streams. Methane leaks and unused hydrogen streams were assessed not only as physical losses, but also as economic loss costs. Methane losses lead to both direct greenhouse gas emissions and lost revenues from energy generation. Although hydrogen losses do not cause any immediate disposal costs, they represent significant opportunity costs. The modelling shows that the use or marketing of hydrogen could generate additional revenues of around €50/t input material. The results thus underline the great importance of efficient gas use for the overall economic efficiency of the system. The investment cost analysis also shows that the fermenter, including agitation technology, accounts for the largest share of capital tied up at around 40%. The combined heat and power plant represents the second largest cost block at around 25%. Although membrane systems and gas treatment have a comparatively small share of the total costs, they have a significant influence on the process yield and the level of loss streams. Investments in these plant components can therefore have disproportionate effects on overall economic performance. The total costs in all countries are in a narrow corridor between around 330 €/t and 340 €/t input material. In contrast to the cost side, the revenue side varies considerably. Revenues range from only approx. 32 €/t in South Africa and Egypt to approx. 120 €/t in Germany and Italy. This range primarily reflects differences in the electricity price level or in the subsidy architecture. The differences in revenues between the countries amount to up to around 90 €/t. In comparison, the maximum difference in total costs is about 10 €/t. This asymmetrical dispersion clearly shows that profitability is not shifted via the cost side, but via the revenue side. In economic terms, this means that the system is not a cost-optimizable system in the classic sense, but a market-dependent revenue system. Success depends primarily on the price at which the energy generated can be marketed. The difference between total costs and revenues results in the economic result per ton of input material. In none of the countries under consideration is full cost recovery achieved.

Table 2 Cost and revenue structure in € per t SHW (including SHW-costs)

Country / Cost-type (€)	Ger-many	France	Italy	Mo-rocco	Algeria	Egypt	South Africa
Material with SHW	300,26	302,76	301,76	300,08	300,07	300,26	300,26
Energy	18,04	18,87	19,36	14,66	17,91	9,66	9,77
System	19,64	19,64	19,64	19,64	19,64	19,64	19,64
Waste	0,26	0,26	0,26	0,26	0,26	0,26	0,26
Total	338,2	341,53	341,02	334,64	337,88	329,82	329,93
Reve-nues	119,75	98,33	122,82	73,54	101,4	32,47	31,62
Profit	−218,45	−243,20	−218,20	−261,1	−236,48	−297,35	-298,31
Material Leaks	53,3	53,3	53,3	53,3	53,3	53,3	53,3

Even in the best-case scenario (Italy), revenues only cover around 36% of total costs. In Egypt and South Africa, the coverage ratio is less than 10%. These figures make it clear that the profitability gap is structurally large and cannot be closed by marginal optimizations. The negative results range from -218 €/t to -298 €/t. The difference between the best and the worst scenario is therefore around 80 €/t. The plant remains clearly loss-making under the assumptions made. Thus, even strongly developed subsidy mechanisms are not sufficient to generate a positive return with high substrate costs. Since the revenue side depends primarily on the electricity price, this represents the most important external sensitivity parameter. A rough estimate shows that a doubling of product revenues would significantly reduce the deficit but still does not fully cover costs in most countries. In Germany, for example, a doubling of revenues from 119.75 €/t to about 240 €/t would reduce the deficit to around -120 €/t. This makes it clear that even drastic increases in revenues only partially compensate for the substrate costs. In the following, the economic efficiency of biogas plants is considered, which – due to regional and temporal market conditions – can purchase slaughterhouse waste in some cases without or with greatly reduced market costs. This scenario is taken up because although there are many industries which, depending on the legal situation and regional market differences, also purchase slaughterhouse waste and thus compete as customers, the corresponding costs can vary too much from region to region. Therefore, a second scenario without costs for slaughterhouse waste is presented to also represent the most favorable case. This shows that all the systems considered generate income that covers or exceeds both the financing costs of the system and the variable annual operating costs. Due to the higher electricity revenues, profits in Europe are higher overall than in North and South Africa.

Table 3 Cost and revenue structure in € per t SHW (excluding SHW costs)

Country / Cost-type (€)	Ger-many	France	Italy	Mo-rocco	Algeria	Egypt	South Africa
Material without SHW	0,26	2,76	1,76	0,08	0,07	0,26	0,26
Energy	18,04	18,87	19,36	14,66	17,91	9,66	9,77
System	19,64	19,64	19,64	19,64	19,64	19,64	19,64
Waste	0,26	0,26	0,26	0,26	0,26	0,26	0,26
Total	38,2	41,53	41,02	34,64	37,88	29,82	29,93
Revenues	119,75	98,33	122,82	73,54	101,4	32,47	31,62
Profit	81,55	56,80	81,80	38,90	63,52	2,65	1,69
Material Leaks	53,3	53,3	53,3	53,3	53,3	53,3	53,3

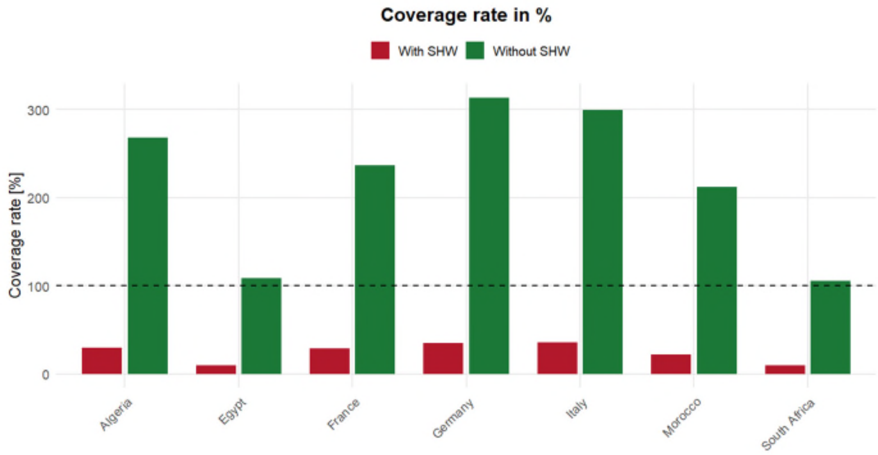

Fig. 6 Coverage rates with and without SHW costs

If one compares the returns of the two very different scenarios, it becomes clear that the economic range of such investments is considerable. In a highly competitive market, there is a minimum loss of at least €218.45 per ton (Germany), while a maximum profit margin of €81.80 per ton (Italy) is possible when purchasing SHW at no cost. Due to a higher proportion of established downstream industries that recycle slaughterhouse waste, it can be assumed that fluctuating prices up to an almost cost-neutral procurement of slaughterhouse waste occur more frequently in

African markets than in Europe. For example, Africa's share of the global market for meat-and-bone meal derived from slaughterhouse waste is around 10-12%, while Europe has market shares of around 30% [11]. For the establishment of biogas plants based on slaughterhouse waste, this results in an uncertain financial viability. This also applies to the African continent, where the animal feed industry based on slaughterhouse waste is developing as a highly competitive industry. Taking into account the determined prices for meat-and-bone meal and tallow, this achieves significantly higher revenues than the energetic use in the context of biogas production [12]. These findings illustrates the importance of a regionally differentiated analysis of market prices and the development of long-term cooperation with the meat industry for the provision of input materials for sustainable electricity generation. In addition, plants should ideally have subsidy mechanisms or contractually secured supplier structures. This is the only way to ensure stable operation with a continuous supply of slaughterhouse waste [13]. For further analyses, the development of a prototype could be used to test the plant costs under practical conditions. Signs of wear, maintenance costs and potential downtimes could be examined in this way. Another important aspect could be to analyze the regional market for slaughterhouse waste and test it as a source of supply to draw conclusions about price developments and possible market fluctuations. In addition to economic security, the importance of operational safety and appropriate state regulation should also be emphasized to avoid both environmental disasters and social risks from the operation of biogas plants.

5. Discussion

The results of the life cycle analysis show that the developed BIOTHEREP system is fundamentally suitable for significantly reducing the negative environmental impacts of the recycling of slaughterhouse waste compared to conventional disposal routes. The environmental benefits are particularly pronounced in countries with fossil-fuelled energy systems, especially in South Africa, Morocco and Algeria. At the same time, it becomes clear that the ecological advantage of such systems is not universal but depends strongly on the respective reference system. This is particularly evident in the example of France, where significantly lower environmental relief is achieved due to the already largely decarbonized electricity mix. The analysis of the modelling showed that, despite its central role in energy generation, fermentation is also the most important environmentally harmful process within the overall system. The results therefore show that although the energetic use of slaughterhouse waste can make an important contribution to reducing emissions, it does not fully compensate for the fundamental environmental impacts of animal production. Additional improvements in environmental performance therefore also require measures along the upstream agricultural value chain.

However, the results of material flow costing show that ecological advantages do not necessarily go hand in hand with economic profitability. Although the technical cost structures are largely comparable between the countries studied, the revenue potential differs considerably. Profitability is determined less by technical parameters than by external market conditions. Electricity prices, support mechanisms for renewable energies and the marketing opportunities for biomethane and other by-products have a significant influence on the economic performance of the system. Of relevance is the consideration of the opportunity costs of slaughterhouse waste. While animal by-products are treated as free residues in many studies, the present MFCA shows that alternative recycling pathways such as the production of animal meals or animal fats can generate considerable economic value. Taking these opportunity costs into account leads to a much more realistic assessment of the economic viability and largely explains the high total cost of the system. At the same time, this result illustrates that the availability of suitable input streams and competition with established recycling channels are decisive factors for the future implementation of such plants. The analysis of the loss flows also shows that comparatively small methane and hydrogen losses have both ecological and economic effects. Methane leaks in particular influence the environmental balance of the system due to the high global warming potential, while hydrogen losses primarily act as a lost economic value. The results therefore underline the importance of high process stability and efficient gas upgrading technologies for the future development of integrated valorisation systems. The sensitivity and uncertainty analyses carried out confirm the robustness of the central results. Despite different weighting schemes and varying model assumptions, the basic rankings of the countries and the identified environmental hotspots remain largely stable. This increases the significance of the results and shows that the observed differences are primarily due to structural properties of the respective energy systems and not to methodological artifacts. The results show that integrated bio- and thermochemical valorisation systems can make an important contribution to the sustainable use of animal by-products. However, in addition to technical improvements, suitable market conditions, long-term raw material availability and supportive political framework conditions are required for large-scale implementation.

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